

ECEN 743: Reinforcement Learning

Value Iteration and Policy Iteration

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References

- [SB, Chapter 3-4]
 - [DB, Chapter 2]
 - [MMT, Chapter 6]
-
- **Acknowledgment:** Some figures for this lecture are taken from UC Berkeley CS188 course, with permission.

MDP Questions

- How do we find the value of a policy π ?

$$V_{\pi}(s) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)\right], \quad a_t \sim \pi(s_t, \cdot), \quad s_{t+1} \sim P(\cdot | s_t, a_t), \quad s_0 = s$$

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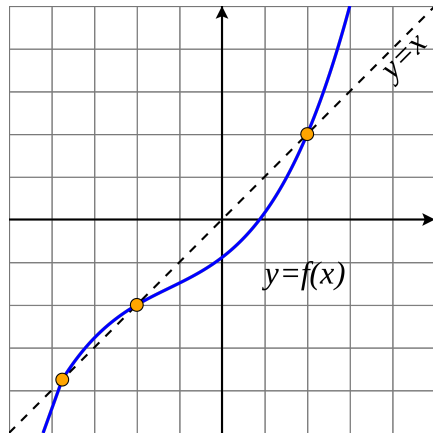
- ▶ Note that $V^* = V_{\pi^*}$

Fixed Point and Contraction Mapping

Fixed Point of a Function

Definition (Fixed Point)

Let $f : \mathcal{U} \rightarrow \mathcal{U}$. u^* is a fixed point of $f(\cdot)$ if $f(u^*) = u^*$.



Contraction Mapping

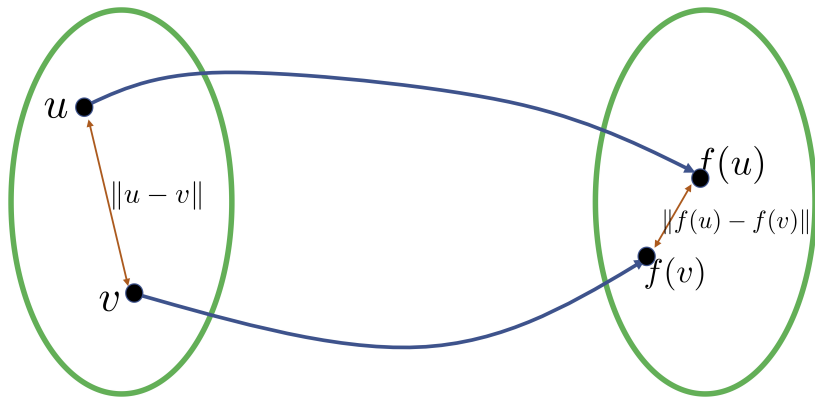
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Let $f : \mathcal{U} \rightarrow \mathcal{U}$ and let $\|\cdot\|$ be a norm defined on $\mathcal{U}$. We say $f(\cdot)$ is a contraction mapping (or simply, contraction) if $\|f(u) - f(v)\| \leq \alpha\|u - v\|$ for an $\alpha \in (0, 1)$ and for all $u, v \in \mathcal{U}$.

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- Example: $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $f(u) = Au$, where $A \in \mathbb{R}^n \times \mathbb{R}^n$. Assume that the row sum of A is strictly less than 1, i.e., $\sum_j |a_{ij}| \leq \alpha < 1$. Then $f(\cdot)$ is a contraction mapping with respect to $\|\cdot\|_\infty$

Banach Fixed Point Theorem

Theorem (Banach Fixed Point Theorem)

Let $\mathcal{U}$ be a closed and bounded subset of $\mathbb{R}^n$. Let $f : \mathcal{U} \rightarrow \mathcal{U}$ be a contraction mapping, i.e., $\|f(u) - f(v)\| \leq \alpha \|u - v\|$, $\alpha \in (0, 1)$, for all $u, v \in \mathcal{U}$. Then,

- ❶ *$f(\cdot)$ has a unique fixed point u^**
- ❷ *The iteration $u_{k+1} = f(u_k)$ with an arbitrary initial element u_0 will converge to the unique fixed point, i.e., $\lim_{k \rightarrow \infty} u_k = u^*$*

Banach Fixed Point Theorem: Proof

Proof: **Existence:** Let $u_0 \in \mathcal{U}$ be arbitrary. Define the iteration $u_{k+1} = f(u_k)$.

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Then, $u^* = \lim_{k \rightarrow \infty} u_k = \lim_{k \rightarrow \infty} u_{k+1} = \lim_{k \rightarrow \infty} f(u_k) = f(\lim_{k \rightarrow \infty} u_k) = f(u^*)$



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So, $\bar{u}^* = u^*$. □

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Theorem (Bellman Optimality Equation)

Let V^ be the optimal value function. Then V^* satisfies the equation*

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- **Reading assignment:** Proof of the above theorem

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- But, this should be the optimal value for state s !

Value Iteration

Bellman Operator

- **Bellman operator** $T : \mathbb{R}^{|S|} \rightarrow \mathbb{R}^{|S|}$ is defined as

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Proposition (Contraction property of Bellman operator)

Bellman operator T is a contraction with respect to $\|\cdot\|_\infty$, i.e, for any $V_1, V_2 \in \mathcal{R}^{|S|}$, $\|TV_1 - TV_2\|_\infty \leq \gamma \|V_1 - V_2\|_\infty$.

A useful result (for the proof)

Lemma

Let g, h be any two real-valued function. Then,

$$\left| \max_a g(x, a) - \max_a h(x, a) \right| \leq \max_a |g(x, a) - h(x, a)|$$

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Proof: (of contraction property)



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$$\begin{aligned} & |(TV_1)(s) - (TV_2)(s)| \\ &= \left| \max_{a \in \mathcal{A}} (r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V_1(s')]) - \max_{a \in \mathcal{A}} (r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V_2(s')]) \right| \end{aligned}$$



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$$\begin{aligned} & |(TV_1)(s) - (TV_2)(s)| \\ &= \left| \max_{a \in \mathcal{A}} (r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V_1(s')]) - \max_{a \in \mathcal{A}} (r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V_2(s')]) \right| \\ &\stackrel{(a)}{\leq} \max_{a \in \mathcal{A}} |(r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V_1(s')]) - (r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V_2(s')])| \\ &= \max_{a \in \mathcal{A}} |\gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V_1(s')] - \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V_2(s')]| \\ &\leq \max_{a \in \mathcal{A}} \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [|V_1(s') - V_2(s')|] \end{aligned}$$



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This implies

$$\|TV_1 - TV_2\|_\infty \leq \gamma \|V_1 - V_2\|_\infty$$



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- **Bellman operator:** $(TV)(s) = \max_a (r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[V(s')])$

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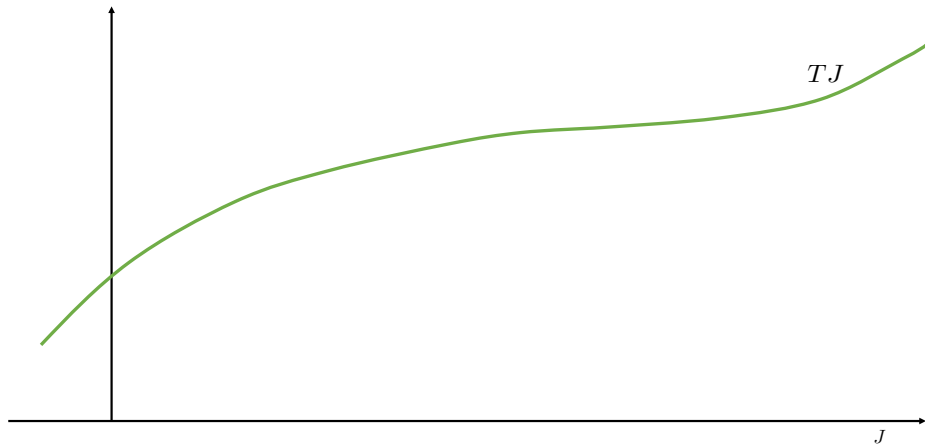
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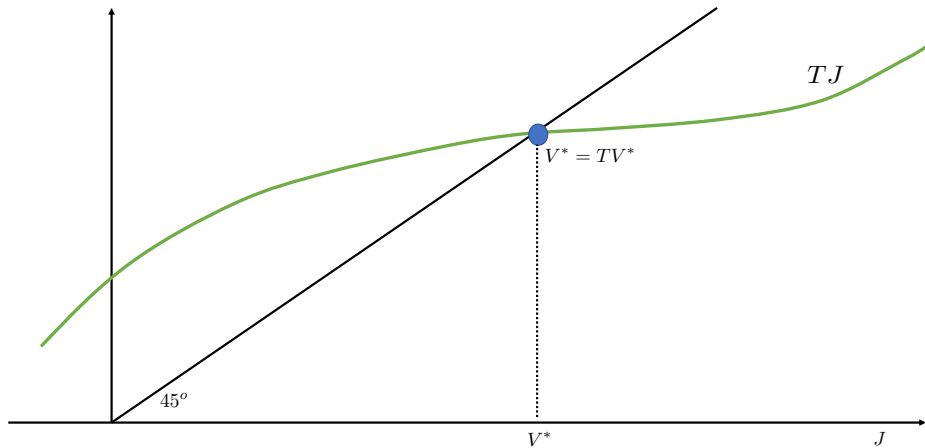
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- Part (2) follows from the fact that T is a contraction mapping and V^* is its unique fixed point

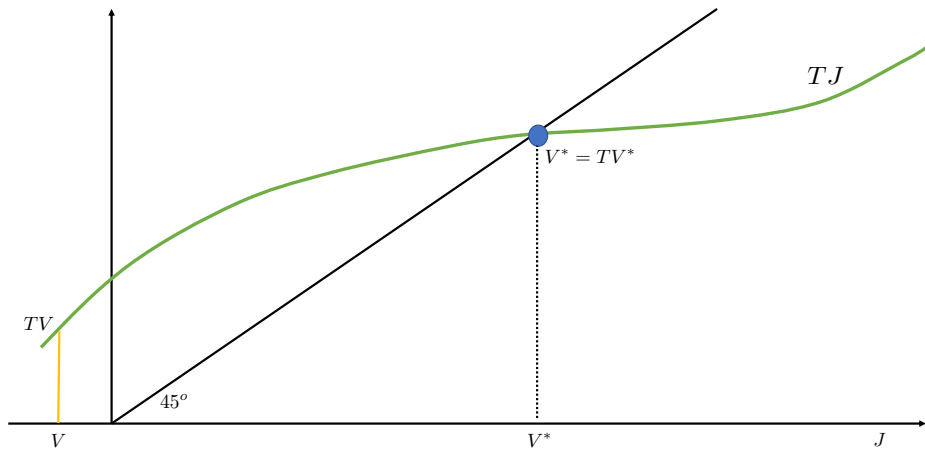
Contraction, Fixed Point, Convergence



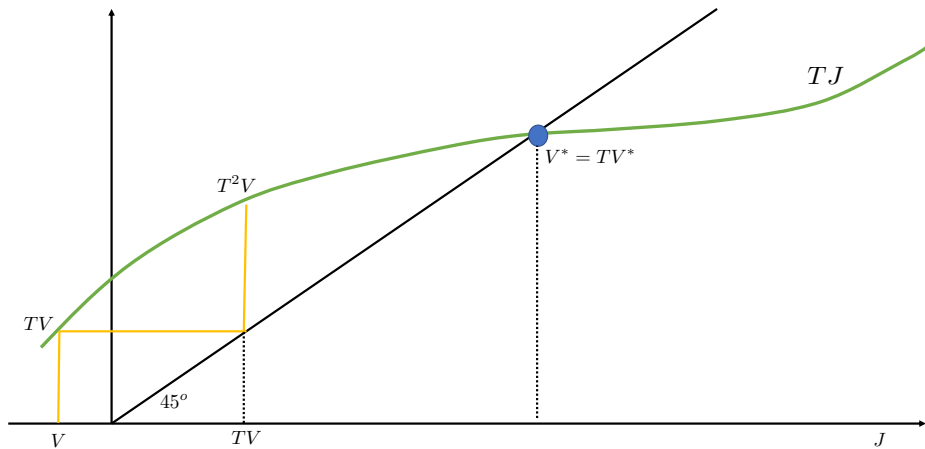
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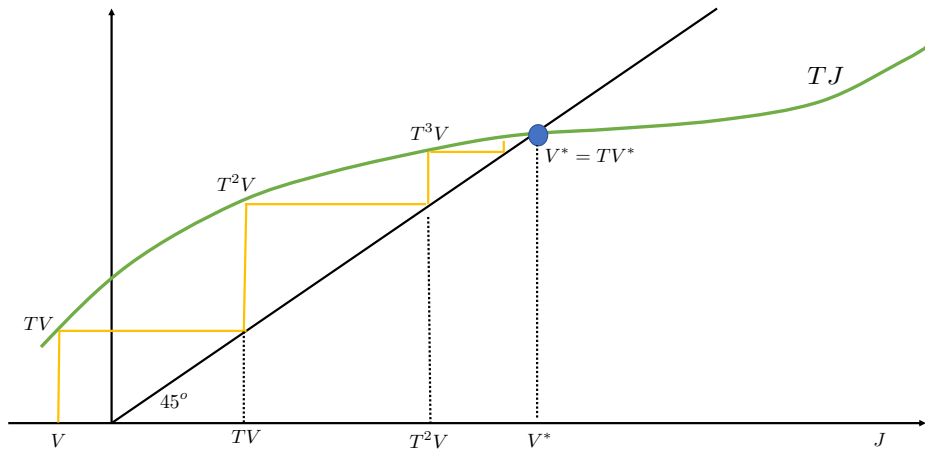
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Value Iteration

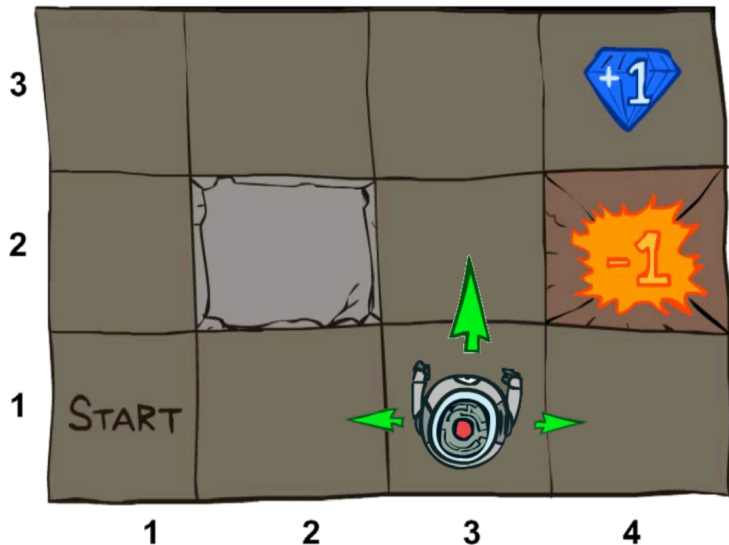
- How do we find the optimal value function V^* ?

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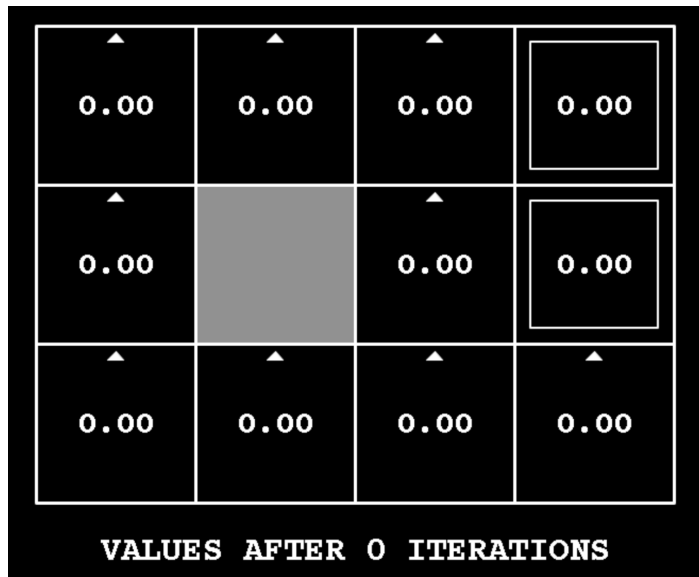
- **Value Iteration:** $V_{k+1} = TV_k$
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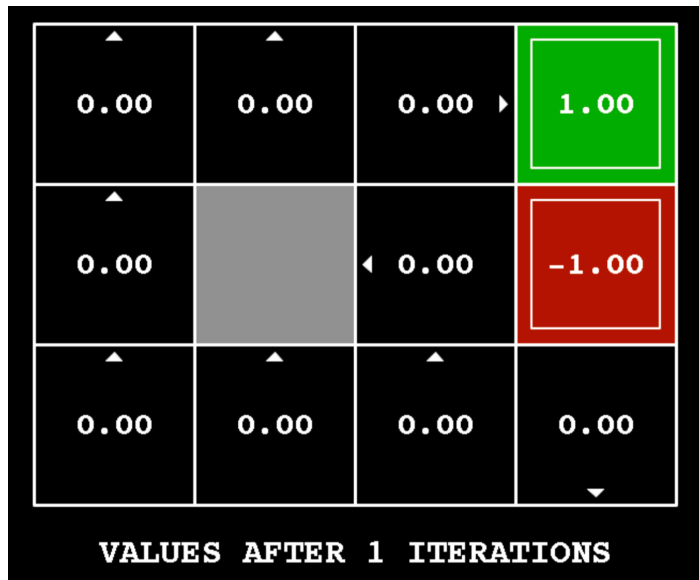
Value Iteration: Example



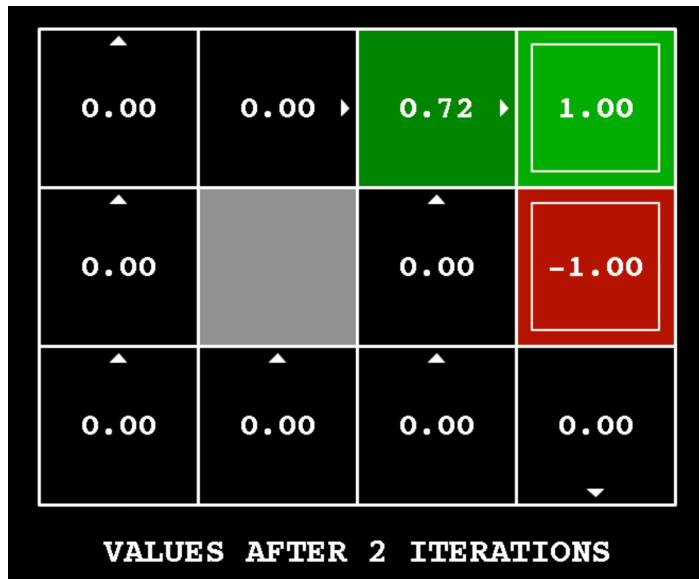
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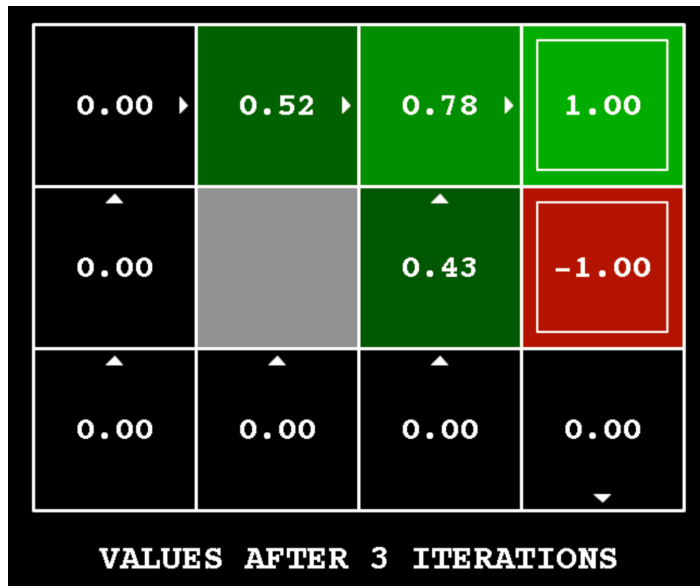
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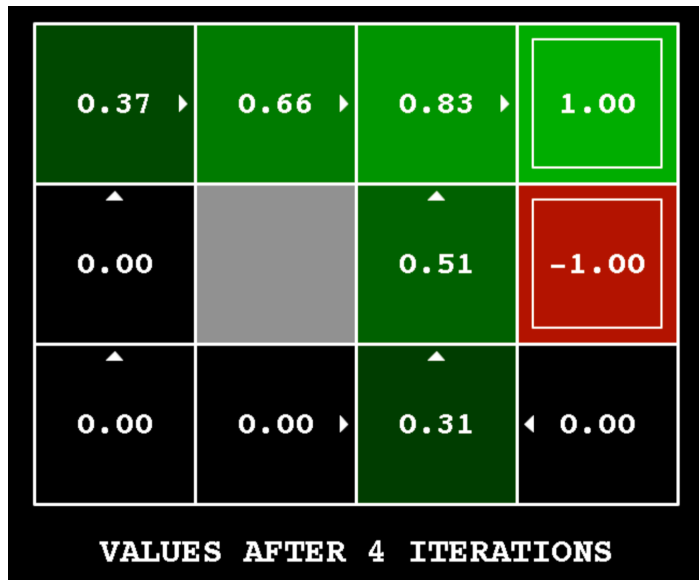
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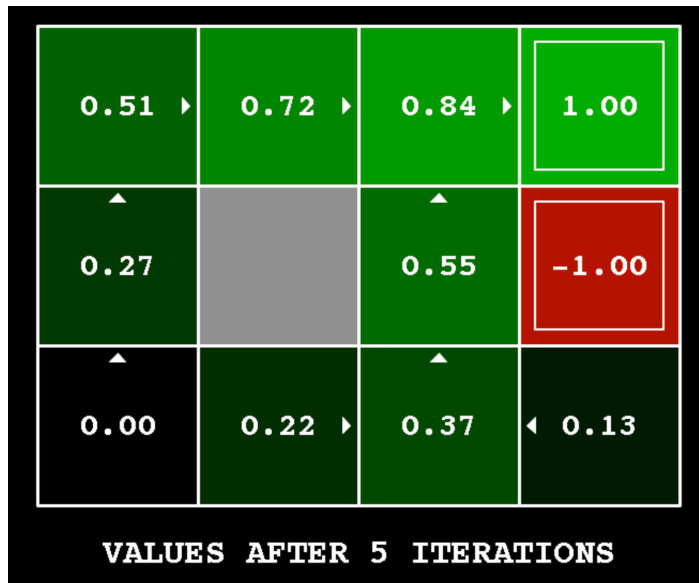
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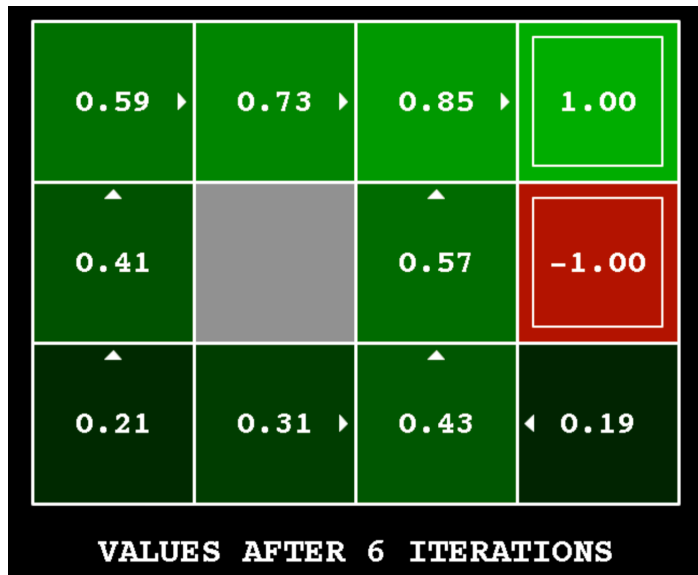
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Optimal Q-value Function

- Q-value function of a policy π

$$Q_{\pi}(s, a) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) | s_0 = s, a_0 = a\right]$$

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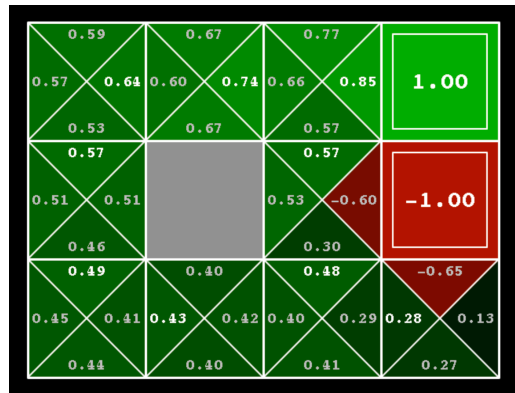
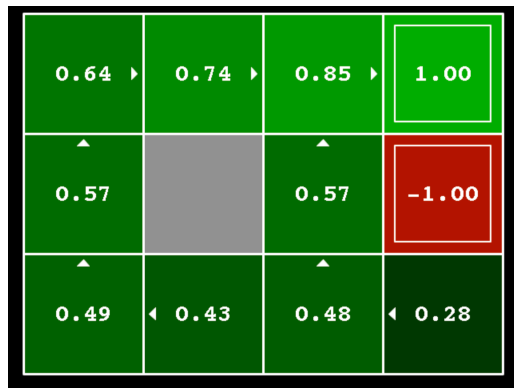
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Policy Computation from V^*/Q^*



$$V^*(s) = \max_a Q^*(s, a), \quad Q^*(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} V^*(s'),$$

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- Optimal Q-value function Q^* satisfies the Bellman equation

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Proposition

- 1 F is a contraction mapping w.r.t $\|\cdot\|_\infty$
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- 3 Define *Q-value iteration*, $Q_{k+1} = FQ_k$. Then, $\lim_{k \rightarrow \infty} Q_k = Q^*$

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Proposition

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- 3 Define the **policy evaluation iteration**, $V_{k+1} = T_\pi V_k$. Then $\lim_{k \rightarrow \infty} V_k = V_\pi$

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Proposition (Convergence of Policy Iteration)

- ① $V_{\pi_{k+1}} \geq V_{\pi_k}$
- ② $\|V_{\pi_{k+1}} - V^*\|_{\infty} \leq \gamma \|V_{\pi_k} - V^*\|_{\infty}$

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Rate of convergence:

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Dynamic Programming

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Dynamic Programming

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 - ▶ Policy evaluation iteration
- How do we find the optimal value function V^* ?
 - ▶ Value iteration
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Dynamic Programming

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- How do we find the optimal value function V^* ?
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 - ▶ Value iteration, policy iteration

Reinforcement Learning Questions

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... without the knowledge of the model P